Beyond Masked and Unmasked: Discrete Diffusion Models via Partial Masking

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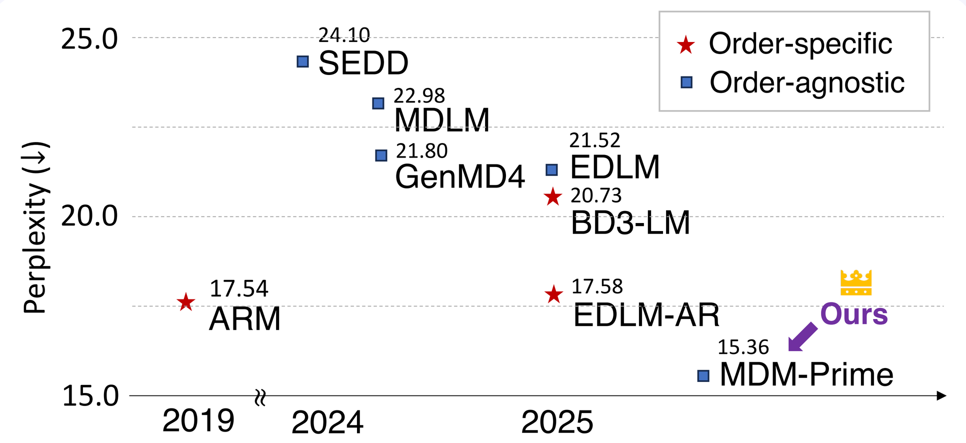
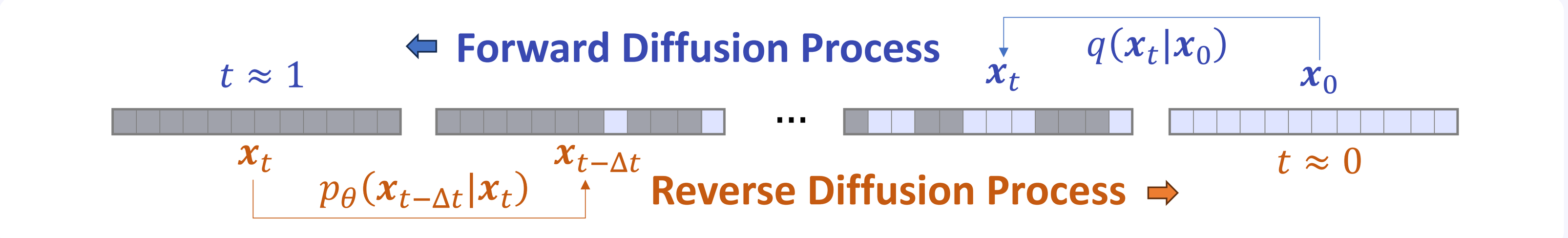
TL'DR

- Standard MDM introduces 37% redundant compute on avg
- Partial masking enables finer denoising → less compute waste
- MDM-Prime establishes the 1st diffusion > autoregressive models

Github **Website**

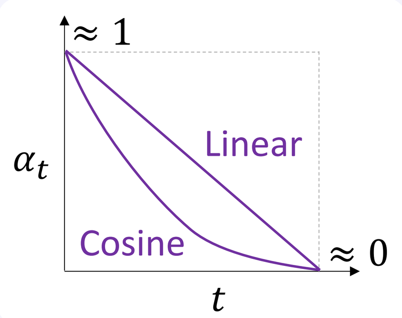


Masked Diffusion Models (MDM)



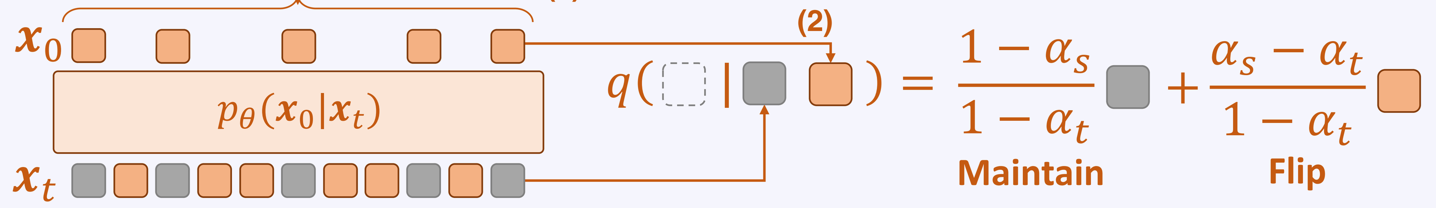
- **Forward Process**

$x_t \sim q(\cdot \mid x_0)$ $q(\cdot|\cdot) = (1-\alpha_t)$ [Flip] $+\ \alpha_t$ [Maintain]



- **Reverse Process**

$$x_{t-\Delta t} \sim p_\theta(\cdot \mid x_t) = \mathbb{E}_{p_\theta(x_0|x_t)}[q(\cdot \mid x_t, x_0)]$$



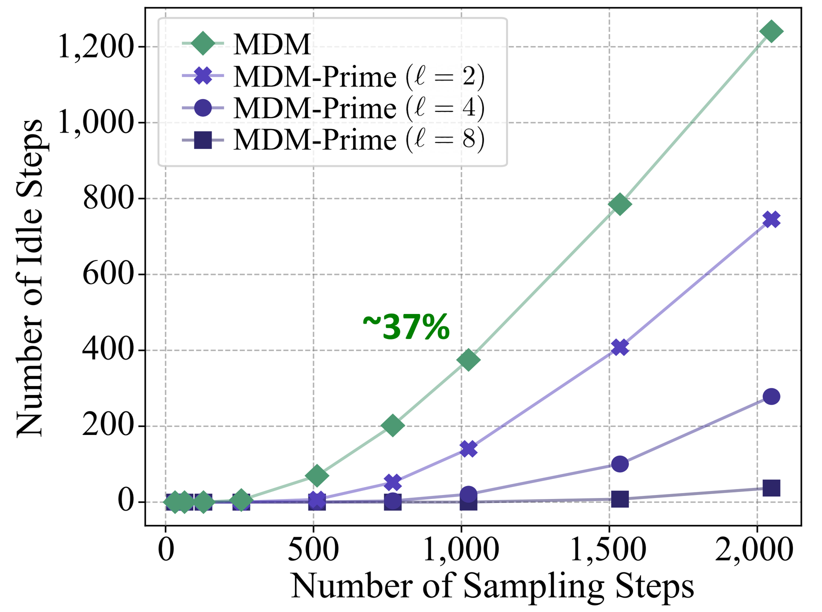
- **Objective**

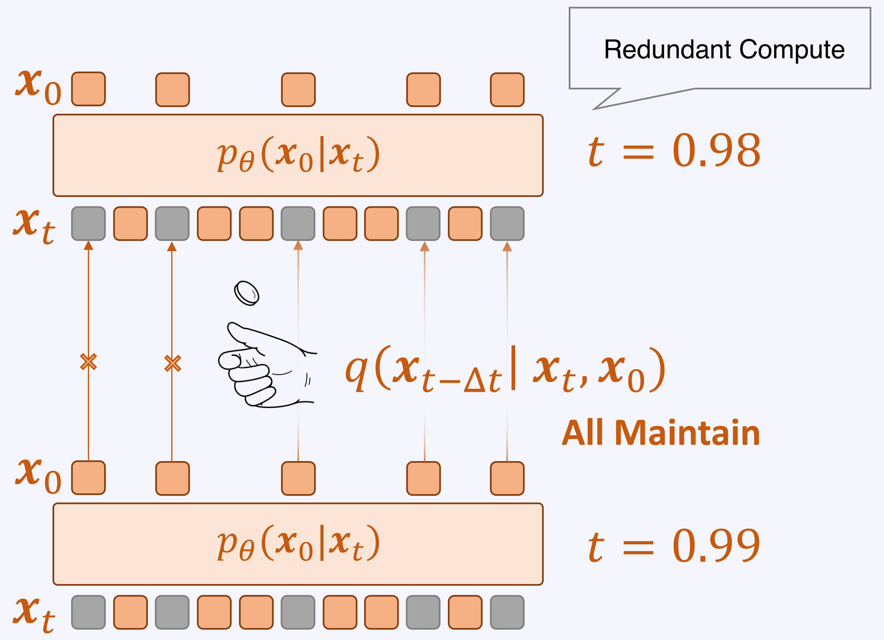
$$-\log p_\theta(x_0) \le L_{vb}(x_0;\theta) = \int_0^1 \underbrace{\frac{\alpha_t'}{1-\alpha_t}}_{\text{Weighting}} \mathbb{E}_{q(x_t|x_0)}\underbrace{\left[\sum_{i=1}^{L} \log p_\theta(x_0^i|x_t)\right]}_{\text{Cross-entropy}} dt$$

Partial Masking Scheme (Prime)

Idle Steps Degrade Model Utilization

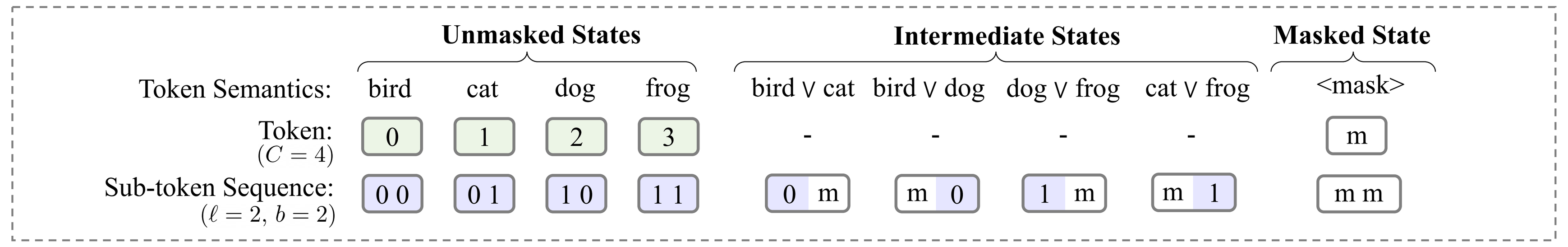
- Sequences remain the same in two consecutive sampling steps → sampling inefficiency

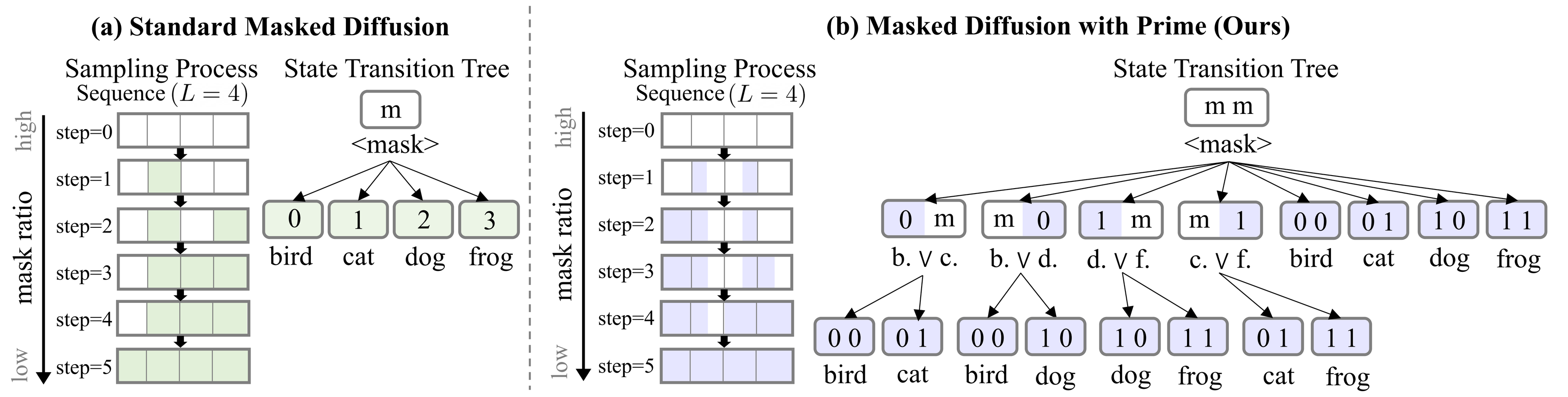




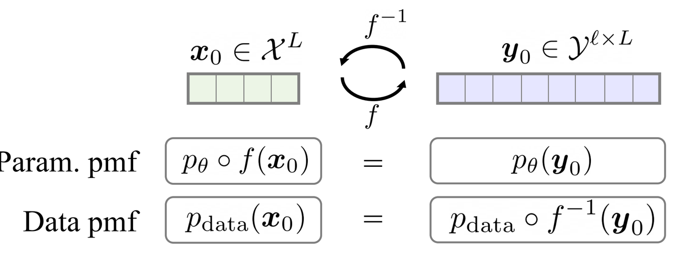
Prime Enables Fine-grained Sampling

- Prime introduces intermediate states → alleviates idle step issue → better model utilization





- 1. Invertible Base-b Encoding



- 2. Diffusion Kernel:

$$q(x_t|x_0) = \prod_{i=1}^{L} q(x_t^i|x_0^i) \to \prod_{i=1}^{L}\prod_{j=1}^{\ell} q(y_t^{i,j}|y_0^{i,j})$$

- (✓) Richer Latent Space
- (✓) Fewer Idle Steps

Implementation

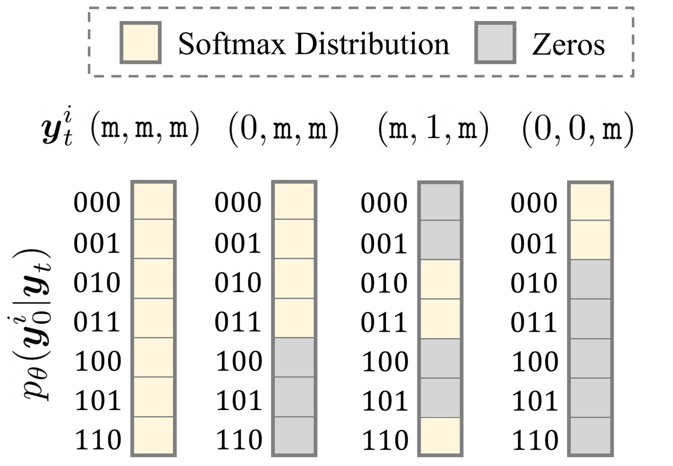
Objective Function

- The objective remains the same after change of variables

$$L_{vb}(y_0;\theta) = \int_0^1 \frac{\alpha_t'}{1-\alpha_t} \mathbb{E}_{q(y_t|y_0)}\left[\sum_{i=1}^{L} \log p_\theta(y_0^i|y_t)\right] dt$$

Efficient Marginal Distribution Calculation

- Marginal distributions can be derived via "carry-over" masks

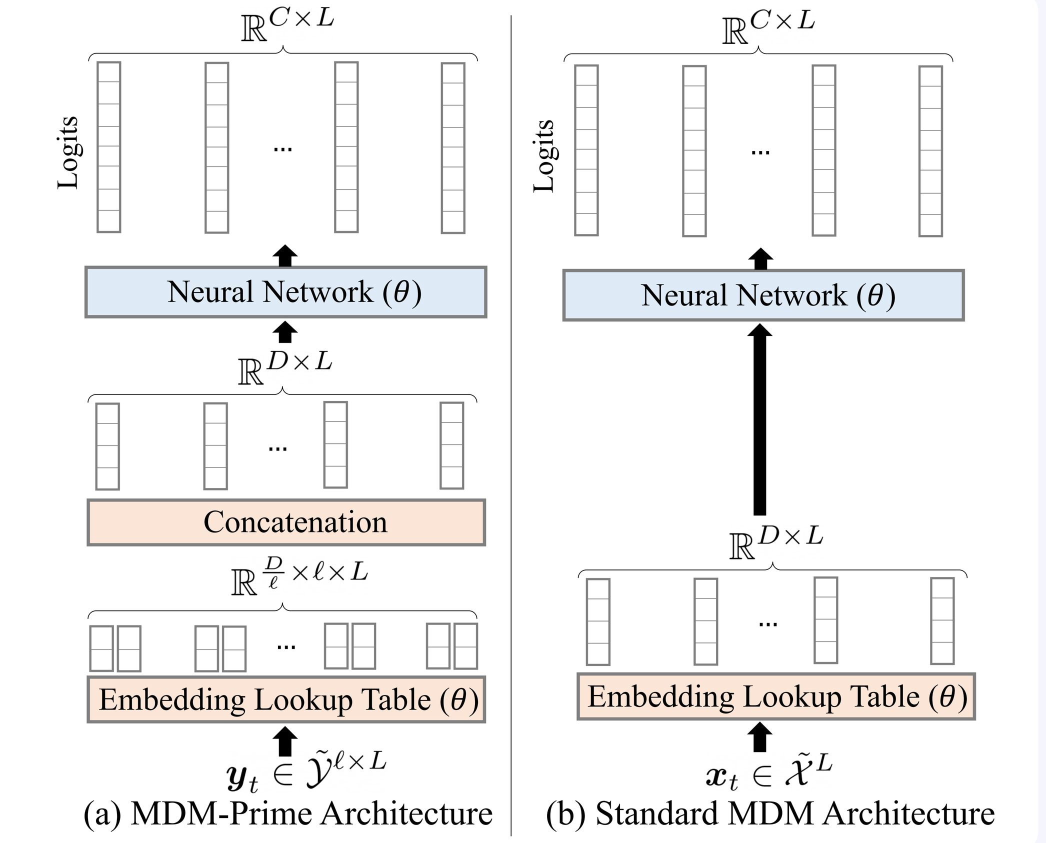


$$p_\theta(y_0^{i,j}|y_t) = \sum_{y_0^{i,1},\dots,y_0^{i,j-1},y_0^{i,j+1},\dots,y_0^{i,\ell}} p_\theta(y_0^i|y_t)$$

$$\triangleq \delta_{y_t^{i,j}}(y_0^{i,j})$$

Simple Architectural Adaptation

- MDM → MDM-Prime by modified embeddings

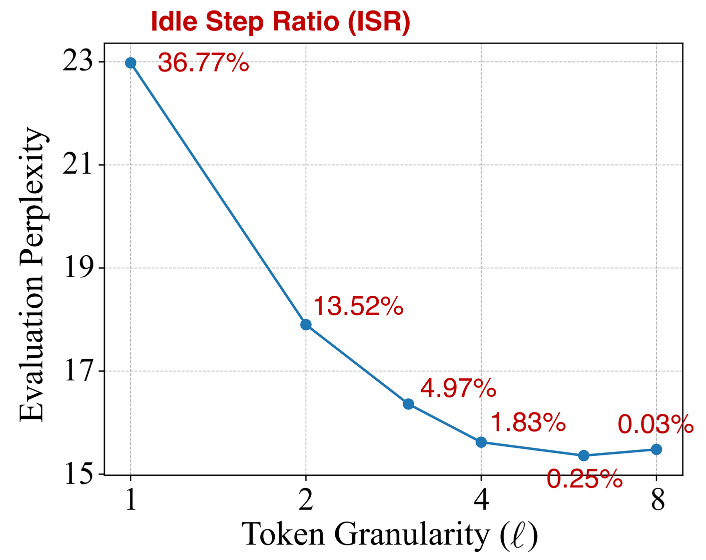


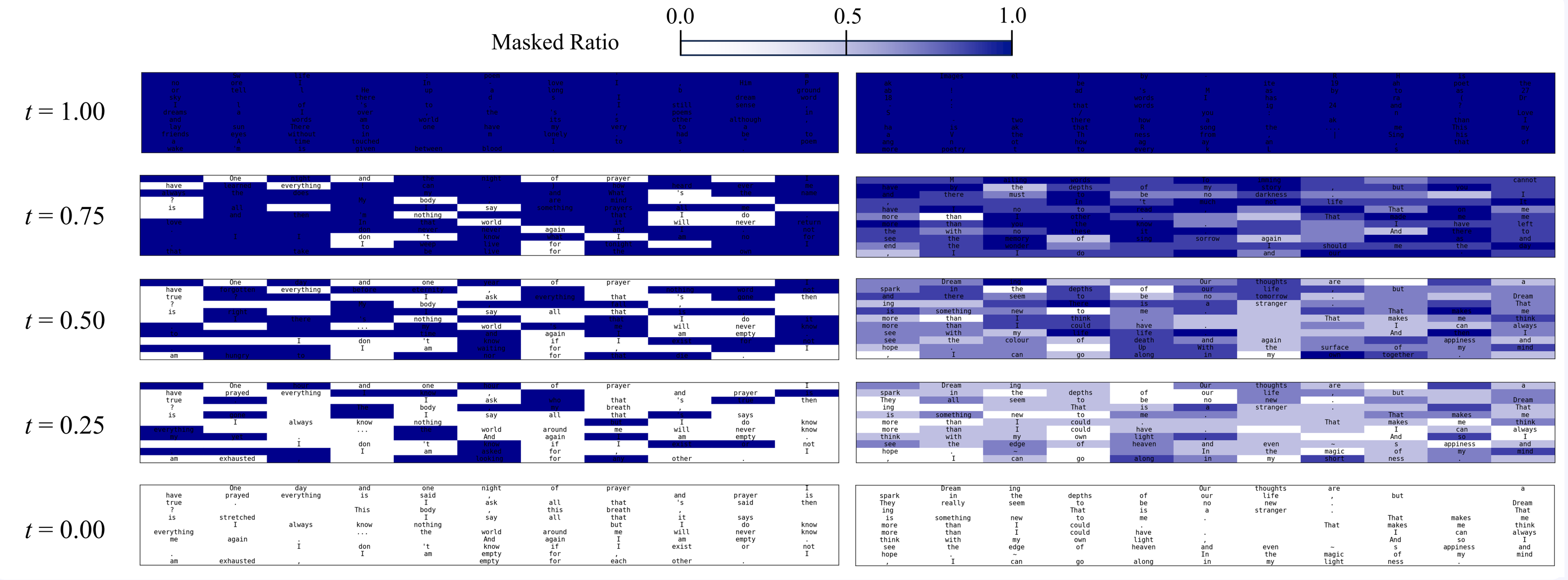
Experiments

- Idle Step Ratio (ISR)
  - e.g., 36.7% for $T = L = 1024$

$$\eta = \left(1 - \frac{1}{T}\right)^{L \times \ell} \quad \eta \downarrow, \ell \uparrow$$

- Scalability w.r.t. Granularity





- Perplexity (PPL)

	PPL (↓)	ISR
ARM* [9]	17.54	-
BD3-LMs* [13]	≤20.73	-
EDLM-coAR* [12]	≤17.58	-
SEDD [7]	≤24.10	-
GenMD4 [10]	≤21.80	-
EDLM-NCE [12]	≤21.52	-
MDLM [9]	≤22.98	36.77%
MDLM-Prime ($\ell = 2$)	≤17.90	13.52%
MDLM-Prime ($\ell = 3$)	≤16.36	4.97%
MDLM-Prime ($\ell = 4$)	≤15.62	1.83%
MDLM-Prime ($\ell = 6$)	≤**15.36**	0.25%
MDLM-Prime ($\ell = 8$)	≤15.48	0.03%

- Zero-shot Evaluation

	LAMBADA	WikiText	PTB	LM1B	AG News	PubMed	ArXiv
ARM* [9]	51.28	25.75	82.05	51.25	**52.09**	49.01	41.73
BD3-LM* [13]	≤50.03	≤31.31	≤96.81	≤60.88	≤61.67	≤42.52	≤39.20
EDLM-coAR* [12]	≤50.04	≤28.31	≤89.73	≤60.23	≤57.94	≤46.31	≤39.02
SEDD [7]	≤49.86	≤34.28	≤100.09	≤68.20	≤62.09	≤41.89	≤38.48
EDLM-NCE [12]	≤46.92	≤30.77	≤93.21	≤63.19	≤60.02	≤41.80	≤36.63
MDLM [9]	≤47.52	≤32.83	≤95.26	≤67.01	≤61.15	≤41.89	≤37.37
MDLM-Prime ($\ell = 2$)	≤30.91	≤27.93	≤74.81	≤55.50	≤63.21	≤**33.32**	≤25.44
MDLM-Prime ($\ell = 3$)	≤27.75	≤27.42	≤60.13	≤42.69	≤62.58	≤41.14	≤**24.71**
MDLM-Prime ($\ell = 4$)	≤**24.44**	≤**23.86**	≤53.98	≤38.02	≤59.44	≤48.64	≤25.83
MDLM-Prime ($\ell = 6$)	≤25.80	≤26.87	≤62.62	≤45.36	≤60.16	≤59.09	≤25.19
MDLM-Prime ($\ell = 8$)	≤25.23	≤25.77	≤**53.77**	≤**38.00**	≤64.89	≤54.50	≤25.79

References

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Acknowledgements